

Project Equations

Overall function : $Y = 5 \times \max(A, B)$

$$A = (A_2, A_1, A_0)_2 \text{ and } B = (B_2, B_1, B_0)_2$$

A_2 and B_2 are MSB's | A_0 and B_0 are LSB's

1. A is greater than B

$$\begin{array}{c} \textcircled{111} > 101 \\ \textcircled{2} \end{array}$$

$$A > B = A_2 \bar{B}_2 + (A_2 B_2 + \bar{A}_2 \bar{B}_2) A_1 \bar{B}_1 + (A_2 B_2 + \bar{A}_2 \bar{B}_2) (A_1 B_1 + \bar{A}_1 \bar{B}_1) \cdot A_0 \bar{B}_0$$

A is less than B

$$A < B = \bar{A}_2 B_2 + (A_2 B_2 + \bar{A}_2 \bar{B}_2) \bar{A}_1 B_1 + (A_2 B_2 + \bar{A}_2 \bar{B}_2) (A_1 B_1 + \bar{A}_1 \bar{B}_1) \bar{A}_0 B_0$$

A is equal to B

$$A = B = (A_2 B_2 + \bar{A}_2 \bar{B}_2) (A_1 B_1 + \bar{A}_1 \bar{B}_1) (A_0 B_0 + \bar{A}_0 \bar{B}_0)$$

$$\therefore (A > B) + (A < B) + (A = B) = 1$$

$$\begin{array}{r} S_2 \ S_1 \ S_0 \\ \hline 3\text{-bit} \end{array}$$

$$\begin{array}{r} S_2 \ S_1 \ S_0 \ 00 \\ \hline 5\text{-bit} \end{array}$$

2. Larger Number Selection

$$\text{NOT gate } (A > B) = \overline{(A > B)}$$

$A > B$	$\overline{(A > B)}$
0	1
1	0

Selection: $S = s_2, s_1, s_0$

$$s_2 = (A > B) A_2 + (\overline{(A > B)}) B_2$$

$$s_1 = (A > B) A_1 + (\overline{(A > B)}) B_1$$

$$s_0 = (A > B) A_0 + (\overline{(A > B)}) B_0$$

$\therefore S = A$ when $A > B$

$S = B$ when $A < B$ or $A = B$

3- Multiplication by 5

$$Y = 5S$$

$$5S = S + 4S$$

$$Y = S + (S \text{ shift left by } 2)$$

	bit5	bit4	bit3	bit2	bit1	bit0
S	0	0	0	s_2	s_1	s_0
$4S$	0	s_2	s_1	s_0	0	0
$5S$	y_5	y_4	y_3	y_2	y_1	y_0

$$y_0 = s_0 ; y_1 = s_1$$

4- Adder Connection's

Adder calculates y_5 to y_2

$$A : A_4 A_3 A_2 A_1 = 000s_2$$

$$B : B_4 B_3 B_2 B_1 = 0s_2 s_1 s_0$$

$$C_0 = 0 / C_{in} = 0$$

5. Output equations:

$$Y_0 = S_0$$

$$Y_1 = S_1$$

$$Y_2 = S_2 \oplus S_0$$

$$= \bar{S}_2 S_0 + S_2 \bar{S}_0$$

$$\text{Carry: } C_3 = S_2 S_0$$

$$Y_3 = S_1 \oplus C_3$$

$$= S_1 \oplus (S_2 S_0)$$

$$\text{Carry: } C_4 = S_1 S_2 S_0$$

$$Y_4 = S_2 \oplus C_4$$

$$= S_2 (\bar{S}_1 + \bar{S}_0)$$

$$Y_5 = S_1 S_2 S_0$$

6. $Y = (Y_5 Y_4 Y_3 Y_2 Y_1 Y_0)_2$

$\downarrow$ MSB $\downarrow$ LSB

$$Y = 5A \text{ when } A > B$$

$$Y = 5B \text{ when } A < B \text{ or } A = B$$

$$\text{Max output: } 5 \times 7 = 35 = (100011)_2$$

3 bit input = $2^3 = 8$.

3 bit input (s)		$3 \times 5 = 15s$						4 s							
decimal value	binary value	decimal value	binary value						decimal value	binary value					
0	000	0	0	0	0	0	0	0	0	0	0	0	0	0	
1	001	5	0	0	0	1	0	1	4	0	0	0	1	0	
2	010	10	0	0	1	0	1	0	8	0	0	1	0	0	
3	011	15	0	0	1	1	1	1	12	0	0	1	1	0	
4	100	20	0	1	0	1	0	0	16	0	1	0	0	0	
5	101	25	0	1	1	0	0	1	20	0	1	0	1	0	
6	110	30	0	1	1	1	1	0	24	0	1	1	0	0	
7	111	35	1	0	0	0	1	1	28	0	1	1	1	0	

(M) $s_2 s_1 s_0$ (L) (M) $o_3 o_2 o_1 o_0 s_1 s_0$ (L) (M) $o_3 s_2 s_1 s_0 o_0 o_0$ (L)

(H) $4s \rightarrow 0 s_2 s_1 s_0 0 0$ (L)

$s \rightarrow 0 0 0 s_2 s_1 s_0$

5s	o_3	o_2	o_1	o_0	s_1	s_0
Output	y_5	y_4	y_3	y_2	y_1	y_0
	M					L

A and B comparators:

$$\cancel{A \neq B} \quad A_2 B_2 + \bar{A}_2 \bar{B}_2 = A_2 \odot B_2$$

$$A_1 B_1 + \bar{A}_1 \bar{B}_1 = A_1 \odot B_1$$

$$A_0 B_0 + \bar{A}_0 \bar{B}_0 = A_0 \odot B_0$$

$$(A > B) = A_2 \bar{B}_2 + A_1 \bar{B}_1 (A_2 \odot B_2) + A_0 \bar{B}_0 (A_2 \odot B_2) (A_1 \odot B_1)$$

$$(A < B) = \bar{A}_2 B_2 + \bar{A}_1 B_1 (A_2 \odot B_2) + \bar{A}_0 B_0 (A_2 \odot B_2) (A_1 \odot B_1)$$

$$\cancel{A = B} \quad (A = B) = (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

A or B selector

if $A > B$ then select A

else if $A < B$ or $A = B$ then select B

so choose ~~B~~ as $A > B$ as selector output G
 $A > B = G$

$$\text{now, } A_2 \cdot G + A_2 B_2 \cdot \bar{G} = S_2 \text{ (MSB of A or B)}$$

$$A_1 \cdot G + B_1 \cdot \bar{G} = S_1$$

$$A_0 \cdot G + B_0 \cdot \bar{G} = S_0 \text{ (LSB of A or B)}$$

multiplication of $\max(A, B)$

⑤ $S = S_2 S_1 S_0$ $S \rightarrow$ selected $\max(A, B)$

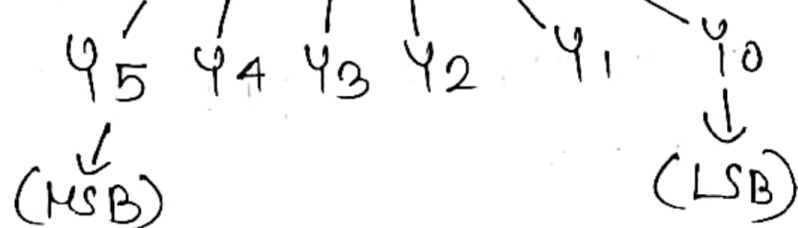
$$5 \times S = 4 \times S + S$$

'in $4 \times S$ bit pattern $\rightarrow 0 S_2 S_1 S_0 0 0$

in S bit pattern $\rightarrow 0 0 0 S_2 S_1 S_0$

So 5s bit pattern $\rightarrow 0_3 0_2 0_1 0_0 S_1 S_0$

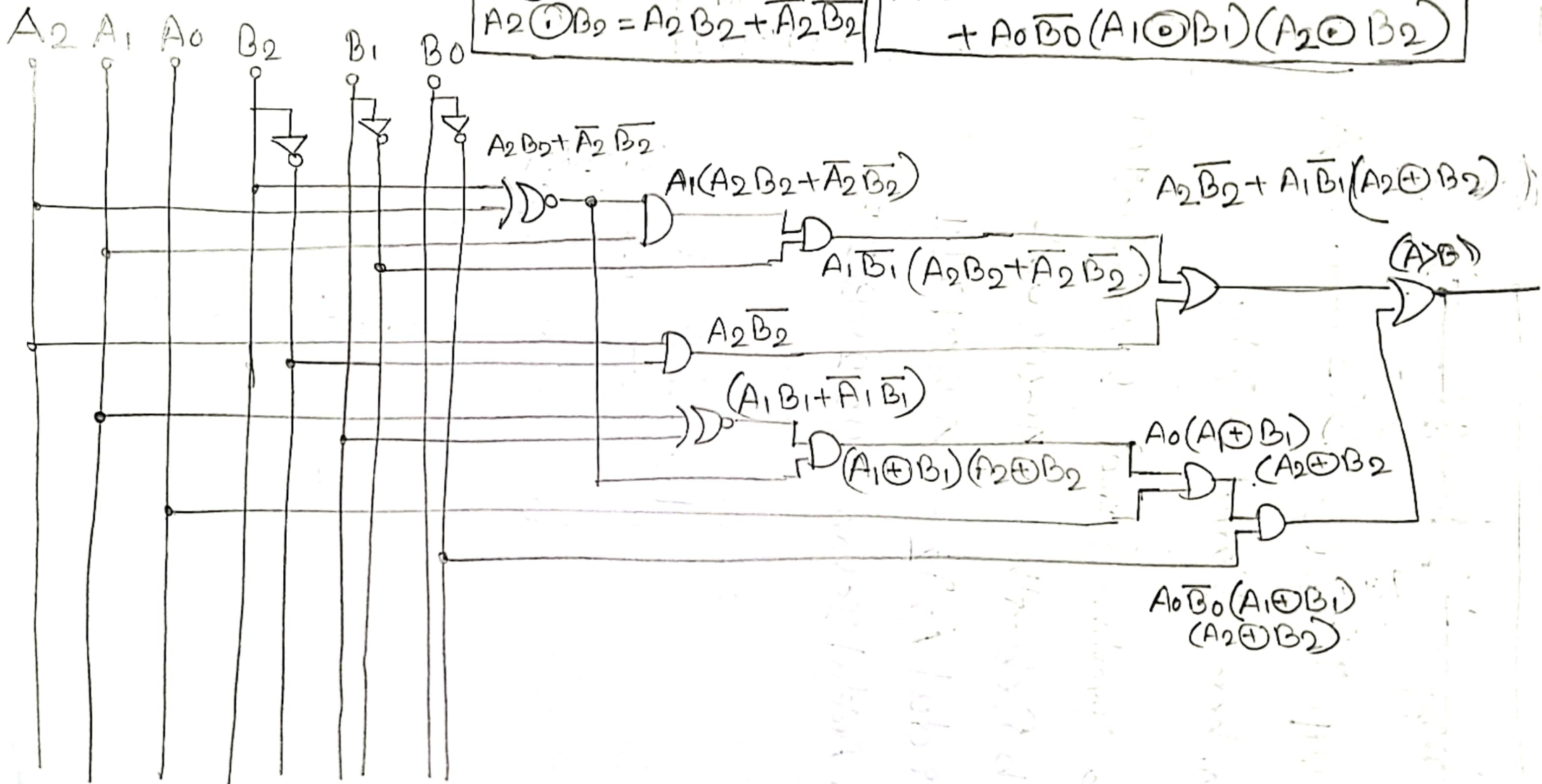
LED $\rightarrow$



$$A_1 \oplus B_1 = A_1 B_1 + \bar{A}_1 \bar{B}_1$$

$$A_2 \oplus B_2 = A_2 B_2 + \bar{A}_2 \bar{B}_2$$

$$A > B = A_2 \bar{B}_2 + A_1 \bar{B}_1 (A_2 \oplus B_2) + A_0 \bar{B}_0 (A_1 \oplus B_1) (A_2 \oplus B_2)$$



A_2 A_1 A_0 B_2 B_1 B_0 $(A > B)$

